

Networking and Web Application step-by-step guide

In this guide we will walk through the process of setting up a secure network infrastructure using Amazon Web Services (AWS) Virtual Private Cloud (VPC) and deploying a simple web application on an EC2 instance.

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Step 1 – Sign in to AWS and Navigate to VPC Dashboard.

- **Sign in** to the **AWS Management Console**, find **“Services”** in the top menu (left side), and directly search and click **“VPC”** as shown in figure 1.

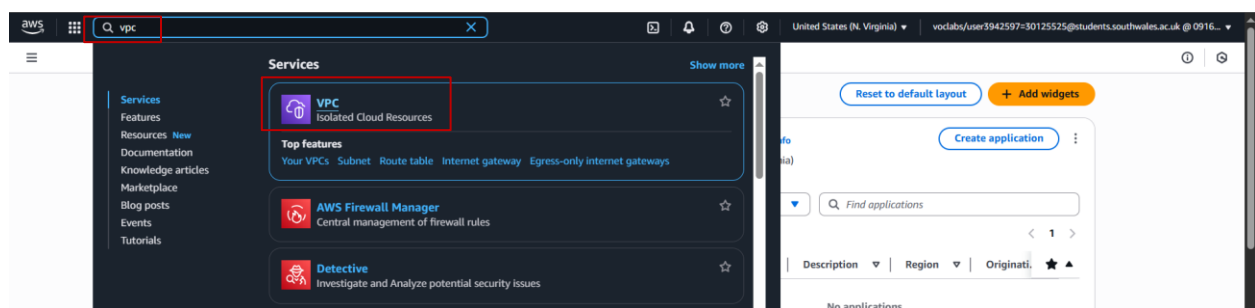


Figure 1. Navigating to VPC Dashboard.

Step 2 – Select AWS Region.

- Select the region you want to work in. Here, we are going to pick **“us-east-1”** as highlighted in figure 2.

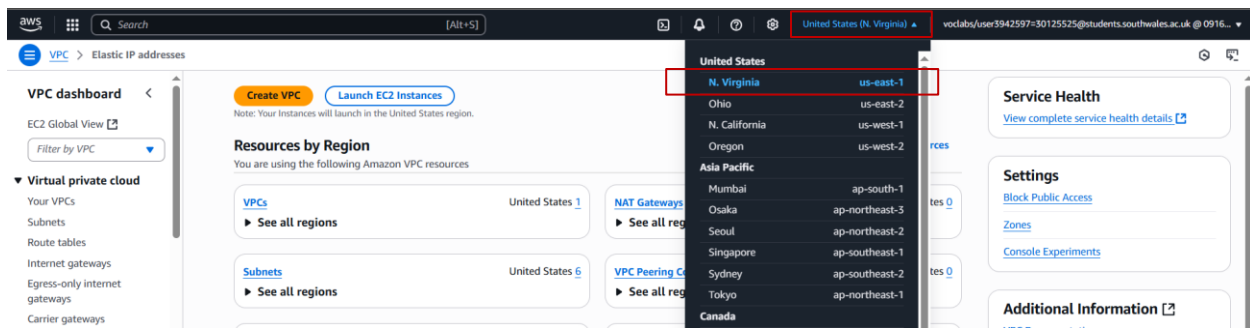


Figure 2. Selecting Region.

Step 3 – Creating a Custom VPC.

- Click **“Create VPC”** to initiate the creating of a VPC as shown in figure 3 below.

It will help DataSafe to control IP addressing, subnets, and routing, which is critical for securing their data and services

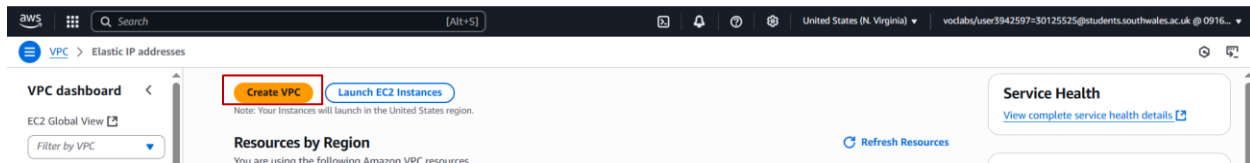


Figure 3. Initiating a VPC.

Step 4 – Naming the VPC.

- Under **“VPC settings”** pick **“VPC only”** as shown in figure 4.
- Under **“Name tag - optional”** you can give your desired name. Here, we are naming it **“30125525-datasafe-vpc”**.

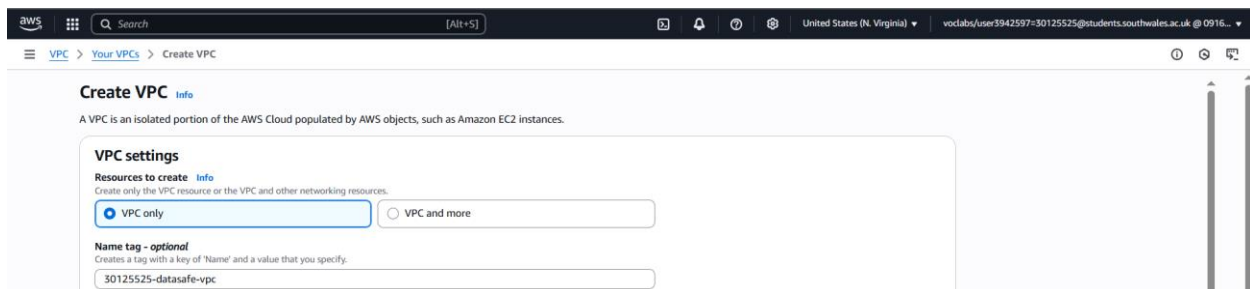


Figure 4. Naming VPC.

Step 5 – Customizing VPC.

- Under “**IPv4 CIDR block**” pick “**IPv4 CIDR manual input**” and under “**IPv4 CIDR**” type “**10.0.0.0/16**” as shown in figure 5.
- Now, under “**IPv6 block**” pick “**No IPv6 CIDR block**” option as highlighted in figure 5.
- “**Tenancy**” should be kept as **Default**.
- Finally click “**Create VPC**” after reviewing all the steps.

Doing these steps defines an isolated network with sufficient IP addresses for subnets.

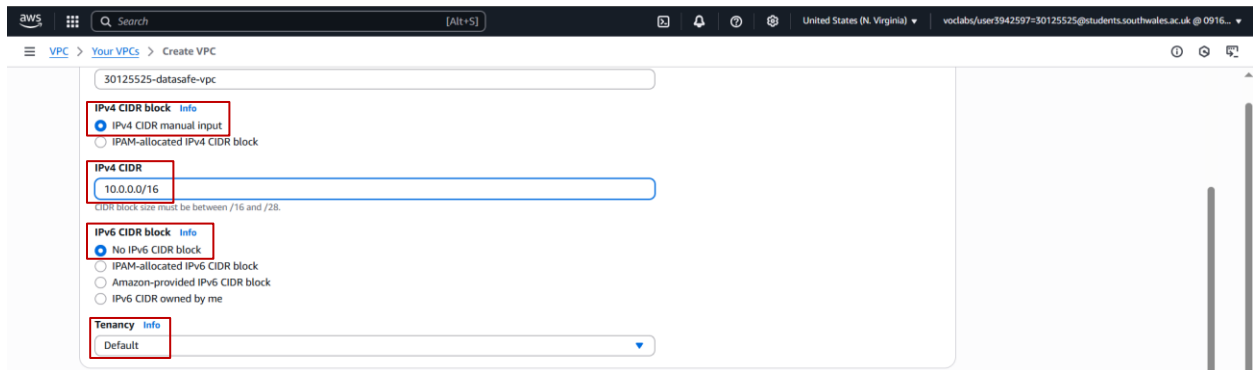
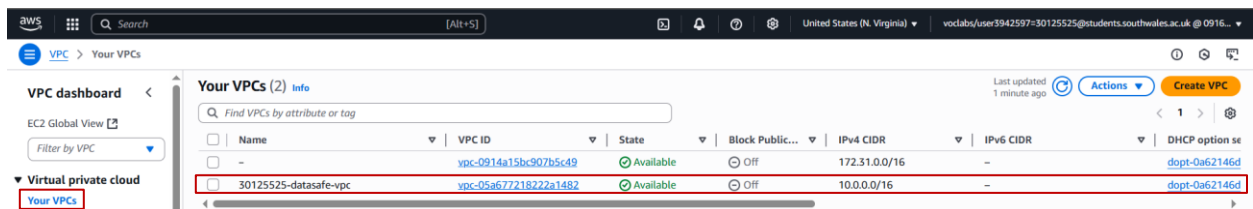


Figure 5. Customizing VPC.

Step 6 – Navigating our newly created VPC.

- On the left pane under “**Your VPCs**” you can see our VPC successfully created and available as shown below in figure 6.



Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP option set
-	vpc-0914a15bc907b5c49	Available	Off	172.31.0.0/16	-	dopt-0a62146d
30125525-datasafe-vpc	vpc-05a677218222a1482	Available	Off	10.0.0.0/16	-	dopt-0a62146d

Figure 6. Successful creation of VPC.

Step 7 – Creating Subnets.

Subnets allow segmentation of the VPC’s IP range into public and restricted areas.

- On the left pane, click “**Subnets**” and then click “**Create subnet**” to start the process of creating a subnet as shown in figure 7.

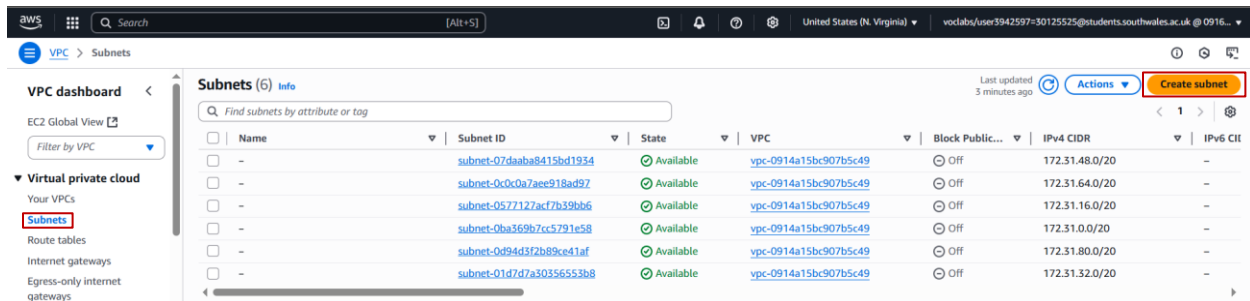


Figure 7. Navigating and creating a Subnet.

Step 8 – Selecting the VPC created above.

- Under “**VPC ID**” dropdown select the VPC created above in step 6 as depicted in the figure 8 below.

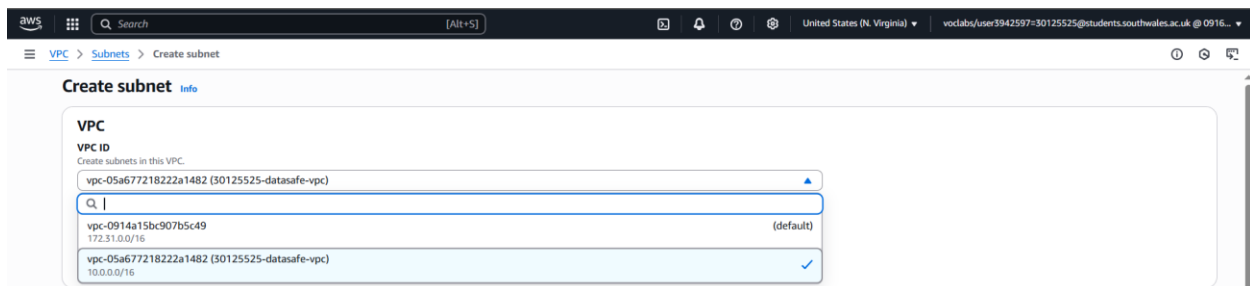


Figure 8. Selecting the VPC created in step 5.

Step 9 – Creating a Public Subnet.

- Under “**Subnet Name**” put the desired name you want. Here, we will name it “**30125525-public-subnet**” as shown below in figure 9.
- Pick “**Availability Zone**” as “**us-east-1a**”.
- For “**IPv4 CIDR block**” type **10.0.1.0/24**.
- Finally, after reviewing click “**Create subnet.**”

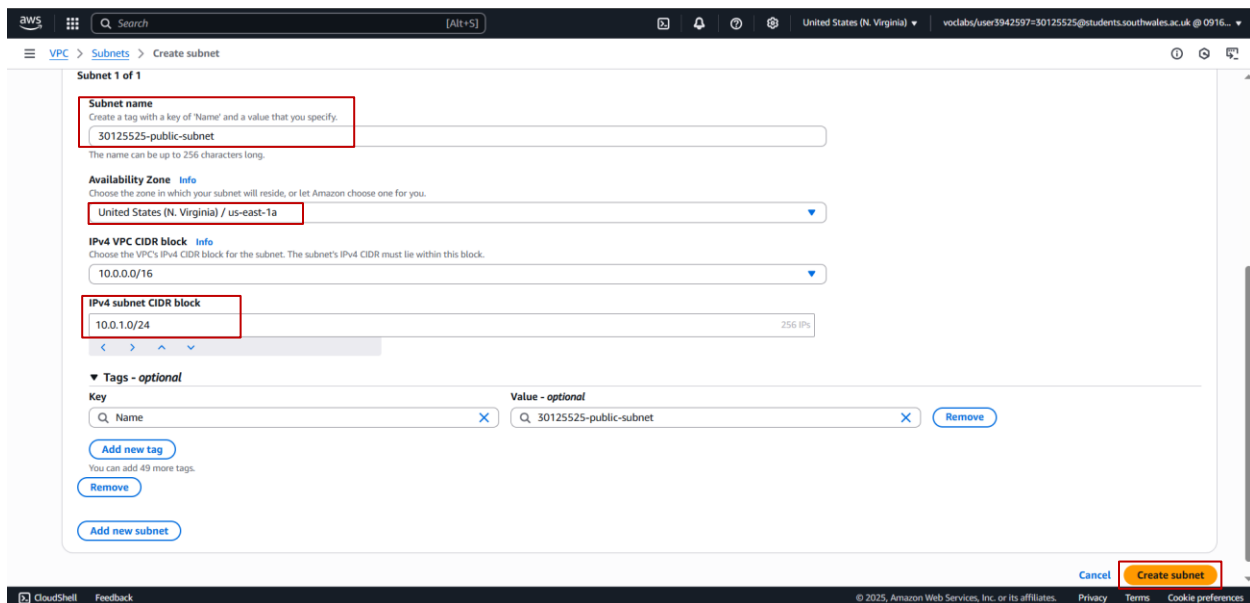


Figure 9. Customizing Public Subnet.

The reason we are creating the public subnet is because it will host the EC2 instance with the web application requiring internet access.

Step 10 – Navigating to our Public Subnet.

- Under “**Subnets**” in the left pane we can see our newly created public subnet as in Figure 10.

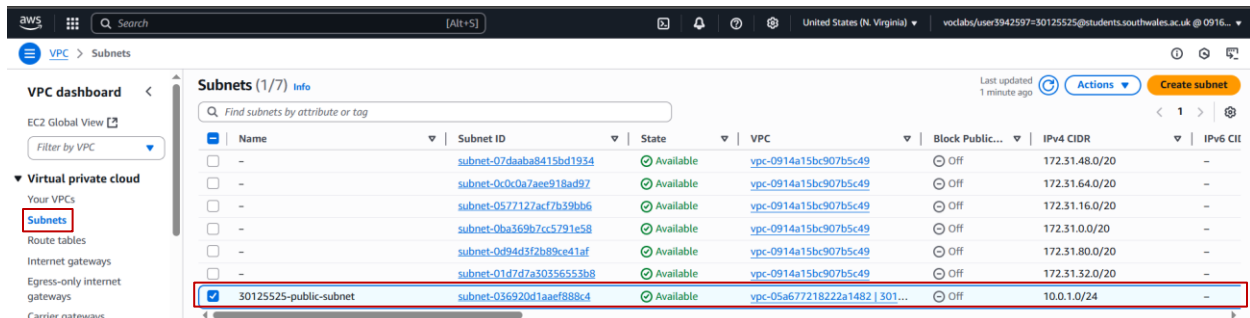


Figure 10. Navigating to our subnet.

Step 11 – Enabling Auto-assign public IPv4 address for public subnet.

- Click “**Subnet ID**” of the newly created public subnet as shown in figure 11.

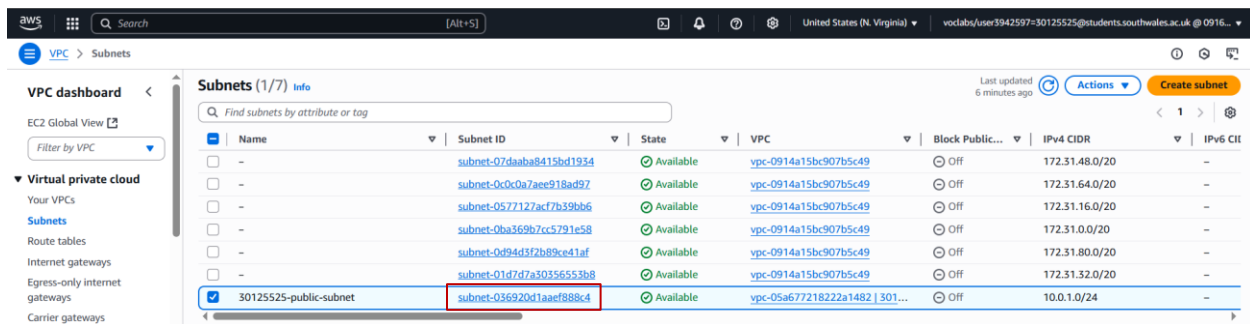


Figure 11. Subnet ID of the newly created public subnet.

- Initially Auto-assign is off, and we need to turn it on. To do so click **“Actions”** under it click **“Edit Subnet settings”** as shown in figure 12.

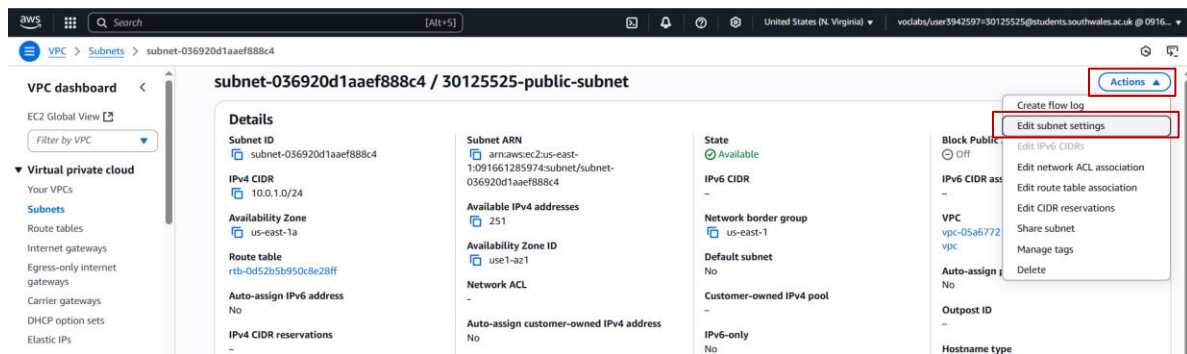


Figure 12. Navigating to subset settings.

- Under **“Auto-assign IP settings”** enable **“Enable auto-assign public IPv4 address”** and click **“Save”** as shown in figure 13.

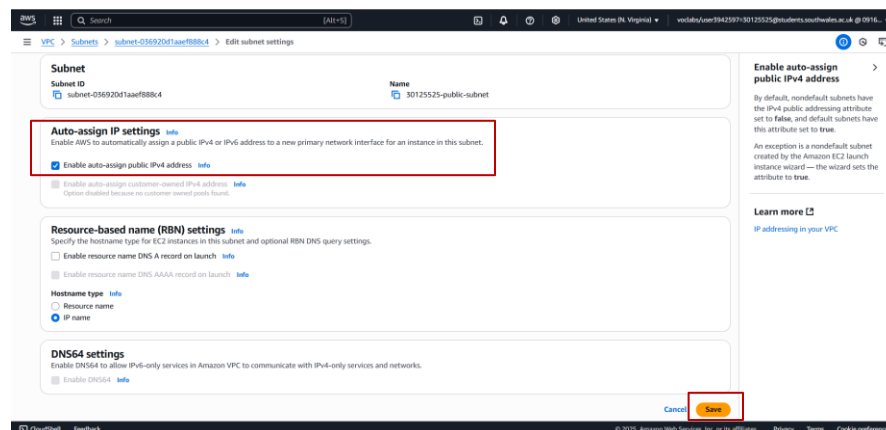


Figure 13. Enabling Auto-Assign public IPv4 address.

Step 12 – Create a Private Subnet.

- Follow **step 7** and **step 8** again.

- Now, under “**Subnet Settings**” write subnet name as “**30125525-private-subnet-1**”.
- Choose “**Availability Zone**” as “**us-east-1b**”.
- For “**IPv4 CIDR block**” type **10.0.2.0/24**.
- All the above steps are highlighted in figure 14.

The screenshot shows the AWS Management Console interface for creating a new subnet. The page is titled 'Subnet 1 of 1'. The 'Subnet name' field contains '30125525-private-subnet-1'. The 'Availability Zone' dropdown is set to 'United States (N. Virginia) / us-east-1b'. The 'IPv4 VPC CIDR block' dropdown is set to '10.0.0.0/16'. The 'IPv4 subnet CIDR block' field contains '10.0.2.0/24'. The 'Tags' section shows a tag with key 'Name' and value '30125525-private-subnet-1'. The 'Add new subnet' button is highlighted with a red box.

Figure 14. Creating the first private subnet.

- Now, we will create another private subnet, to do so click “**Add new subnet**” as shown in figure 14.
- Now, under “**Subnet Settings**” write subnet name as “**30125525-private-subnet-2**”.
- Choose “**Availability Zone**” as “**us-east-1b**”.
- For “**IPv4 CIDR block**” type **10.0.3.0/24**.
- The above steps are highlighted in figure 15.
- After reviewing all the steps click “**Create subnet.**”

The reason we are creating a private subnet is because it contains internal resources unavailable from the Internet, increasing security for any sensitive data or sensitive services hosted. A separate region from the public subnet for the private subnet would improve high availability by separating the resources into different data centers.

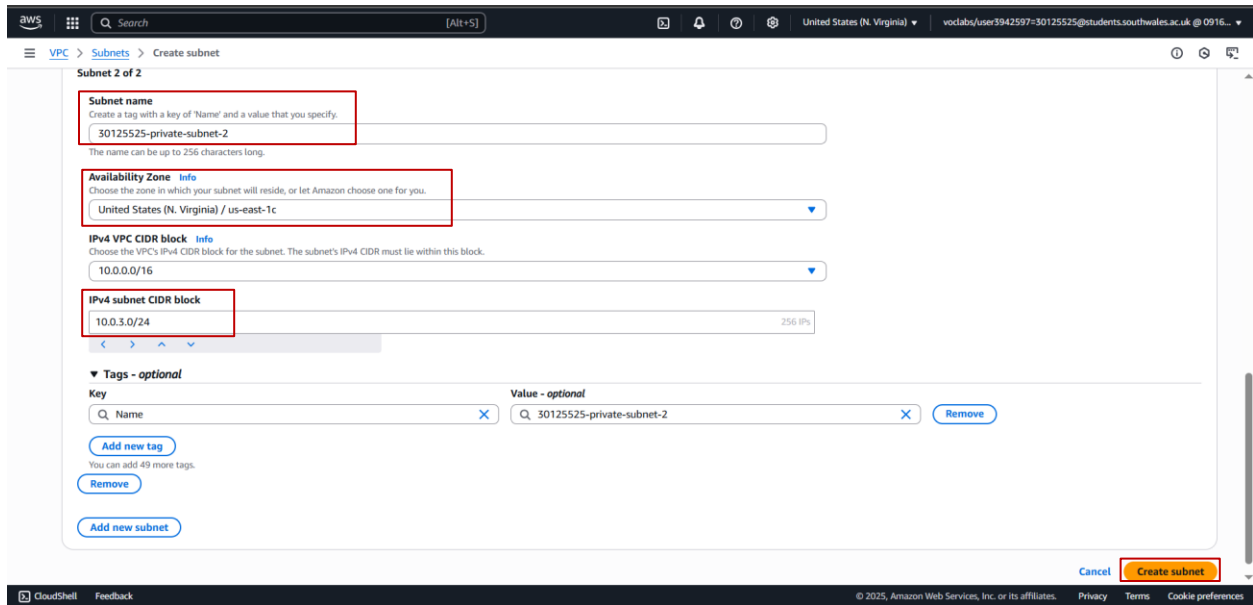


Figure 15. Creating our second Private subnet.

- We can see our two new private subnets are created as shown in figure 16.

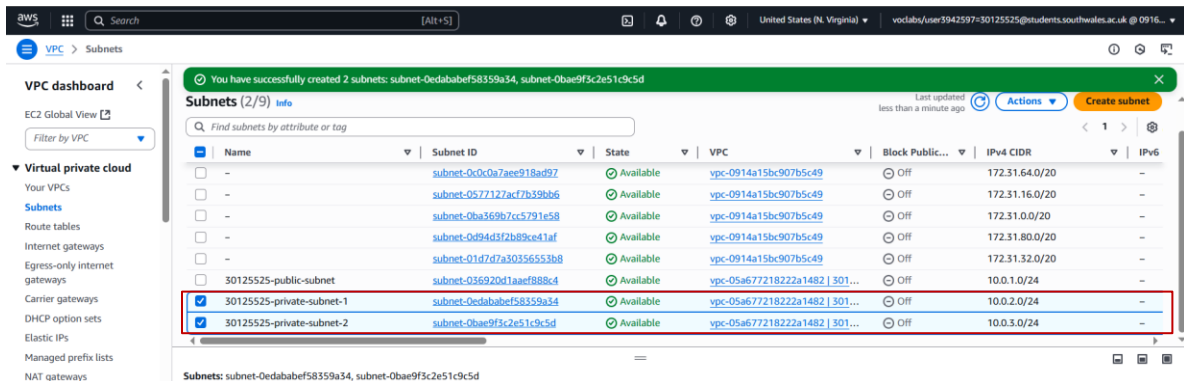


Figure 16. Newly created Private Subnets.

Step 13 – Creating an Internet Gateway.

- From the left pane click “**Internet gateways**” and then click “**Create internet gateway**” as seen in figure 17.

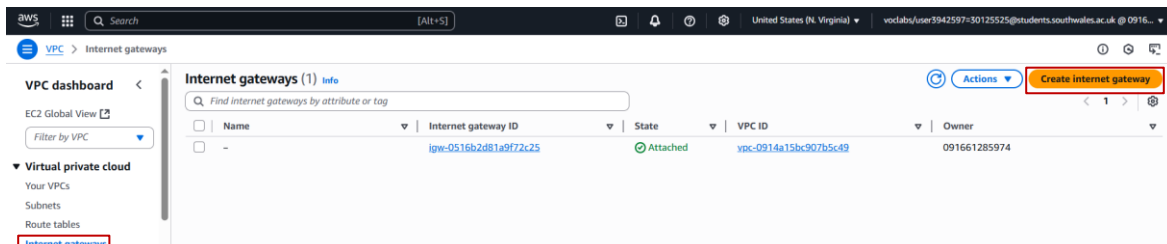


Figure 17. Navigating to create an Internet gateway.

Step 14 – Naming the Internet Gateway.

- Under “**Internet gateway settings**” you can put any name of your choice for easy identification. Here, we will put the name as “**30125525-datasafe-igw**”.
- Click “**Create internet gateway**” to create it as shown in figure 18.

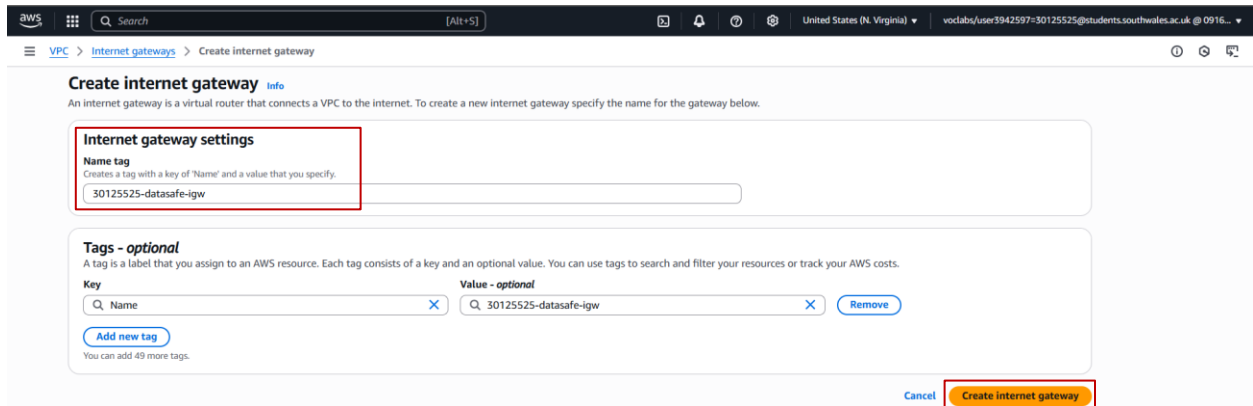


Figure 18. Creating an Internet Gateway.

- We can see the newly created Internet Gateway as shown in figure 19.

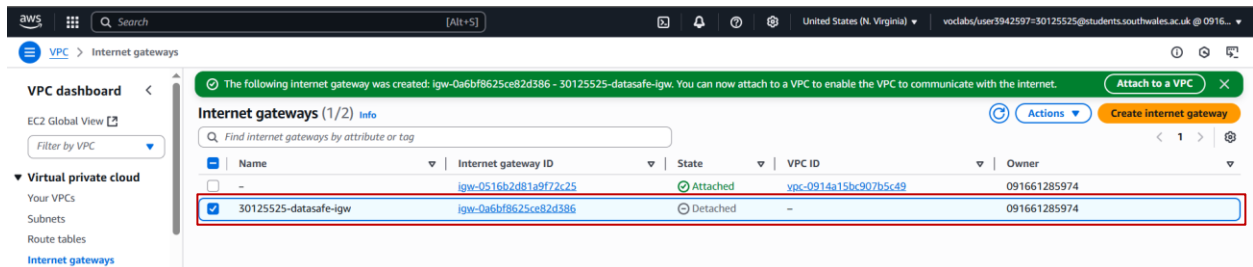


Figure 19. Newly created Internet Gateway.

An Internet Gateway is necessary to allow resources in the public subnet to communicate with the internet

Step 15 – Attaching Internet Gateway to VPC.

- Select “**30125525-datasafe-igw**” and under “**Actions**” click “**Attach to VPC**” as shown in figure 20.

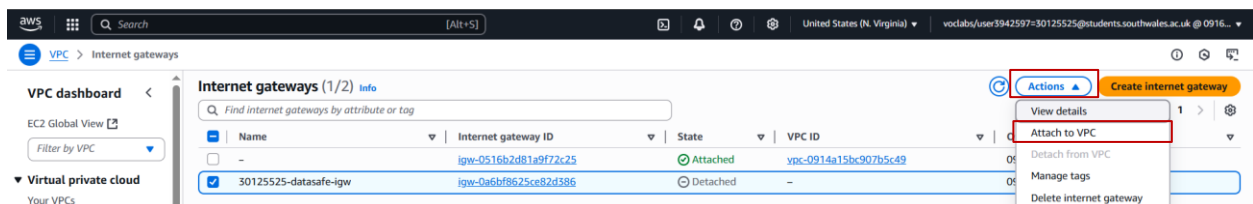


Figure 20. Navigating to “Attach to VPC.”

- Under “**Available VPCs**” select the VPC that we have created before to attach to it as shown in figure 21.

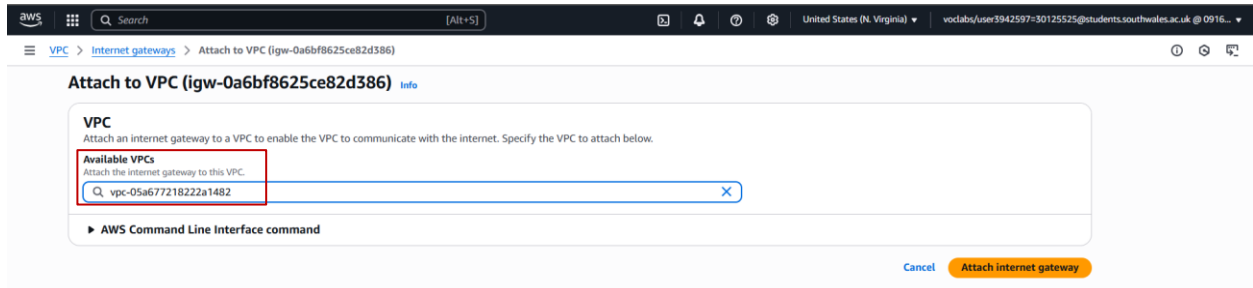


Figure 21. Attaching to a VPC.

- Finally, we can see that our Internet Gateway is attached to our VPC as in figure 22.

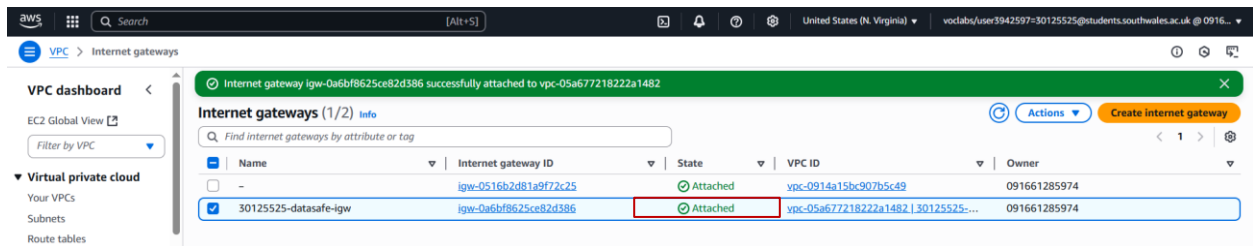


Figure 22. VPC Attached to Internet Gateway.

Step 16 – Creating a route table.

- From the left pane select “Route Tables” and the click “Create route table” as depicted in figure 23.

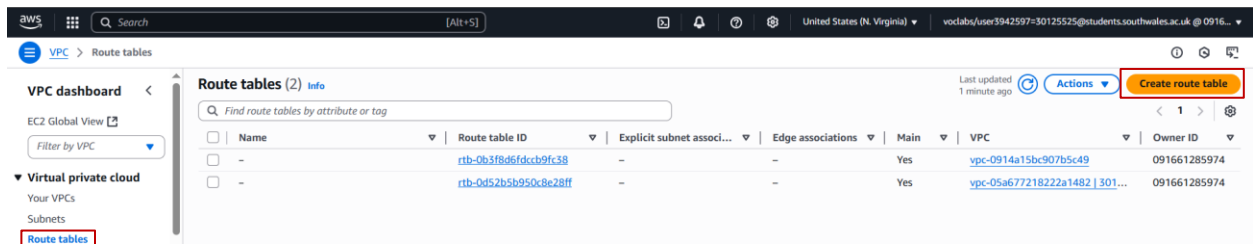


Figure 23. Navigating to Route Table.

- Under “**Route Table Settings**,” put name as “**30125525-public-rt**”.
- Select the “**VPC**” that we recently created in the above steps from the drop-down menu.
- Now, click “**Create route table**” to create one.
- All steps are shown in the figure 24 below.

A route table defines how traffic is routed within the VPC.

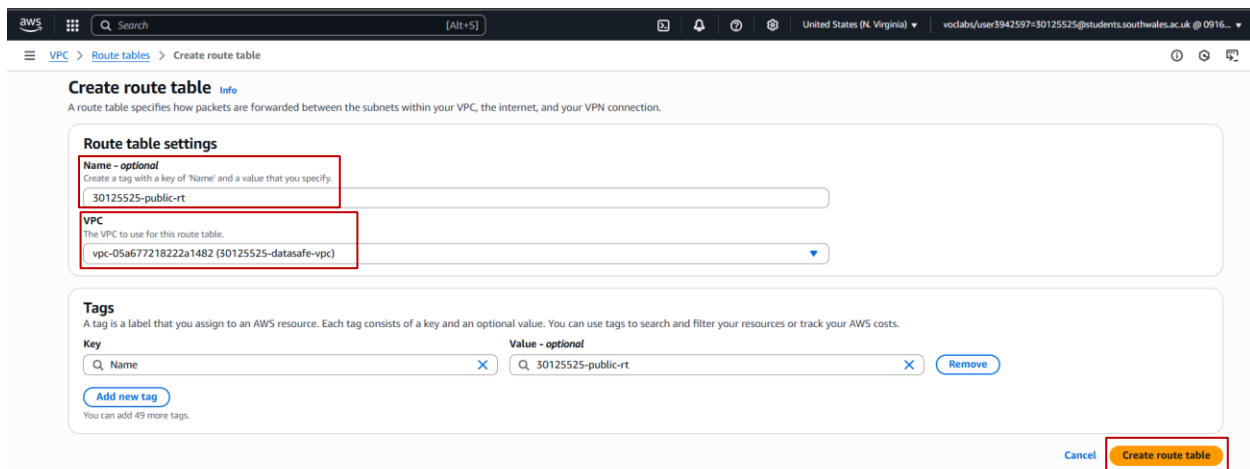


Figure 24. Customizing and creating Route Table.

Step 17 – Adding Route for Internet Access.

- Select the route table and under “**Actions**” click “**Edit routes**” as shown in figure 25.

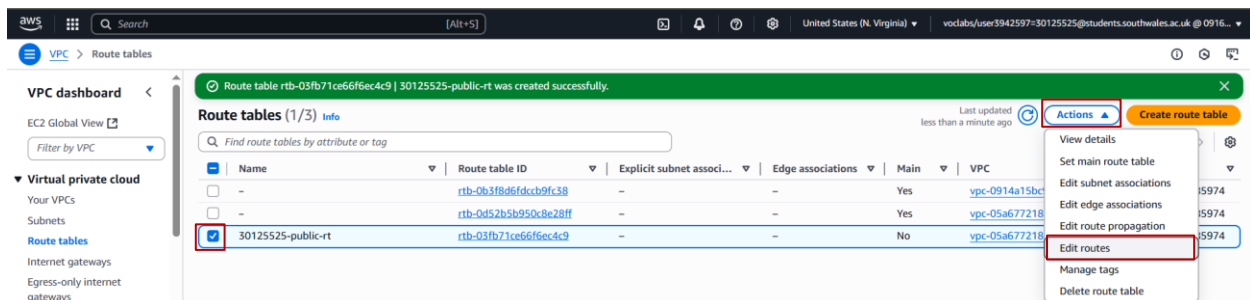


Figure 25. Navigating to edit route.

Step 18 – Customizing Route for Internet Access.

- Click “**Add route**”
- Put Destination as “**0.0.0.0/0**” and select “**Internet Gateway**”.
- From the drop-down menu select our gateway “**30125525-datasafe-igw**” and click “**Save routes**” as shown in figure 26.

The 0.0.0.0/0 destination routes all outbound traffic to the Internet Gateway, allowing the public subnet’s resources, like the EC2 instance, to communicate with the internet.

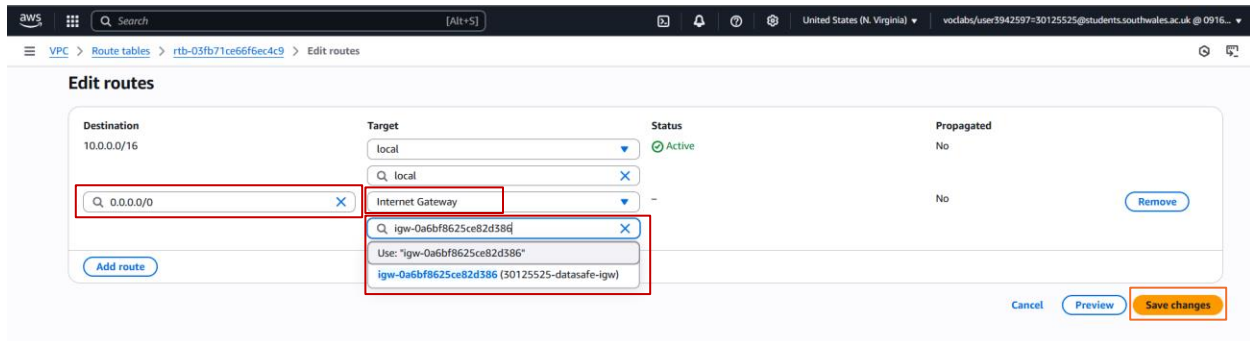


Figure 26. Customizing Routes to our Internet Gateway.

Step 19 – Associating the Route Table to the Public Subnet.

- Select the subnet and go to “**Actions**” and the click “**Edit subnet associations**” as shown in figure 27.

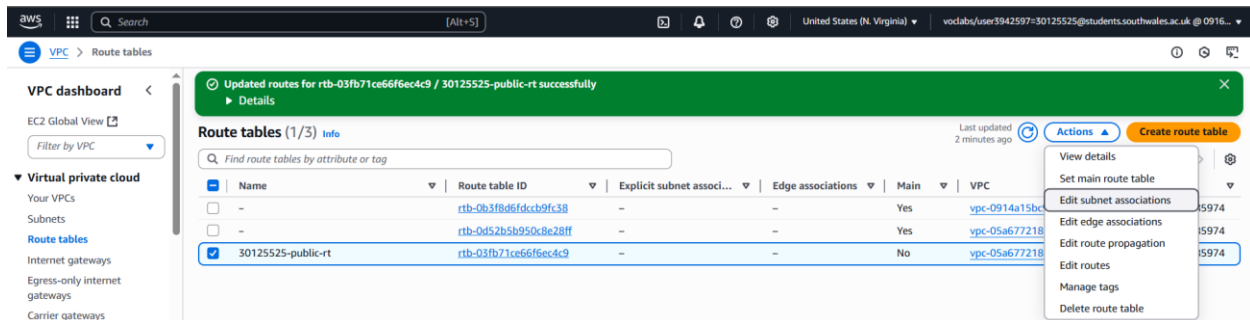


Figure 27. Navigating to edit Route.

- Select our public subnet “**30125525-public-subnet**” and click “**Save associations**” as shown below in figure 28.

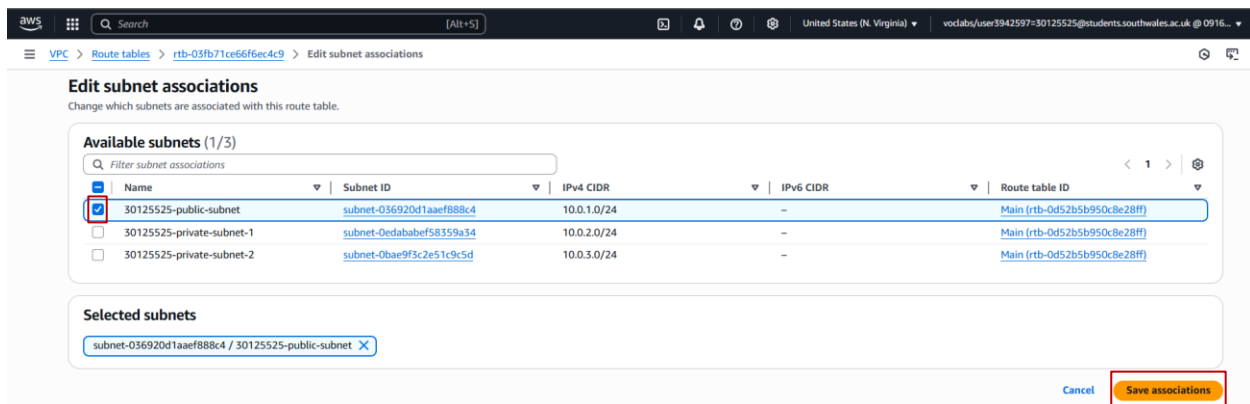


Figure 28. Associating Route to Public Subnet.

By associating the route table with the public subnet, instances in that subnet can send and receive internet traffic, making the web application public.

Step 20 – Sign in to AWS and Navigate to EC2 Dashboard.

- **Sign in** to the **AWS Management Console**, find **“Services”** in the top menu (left side), and directly search and click **“EC2”** as shown in figure 29.

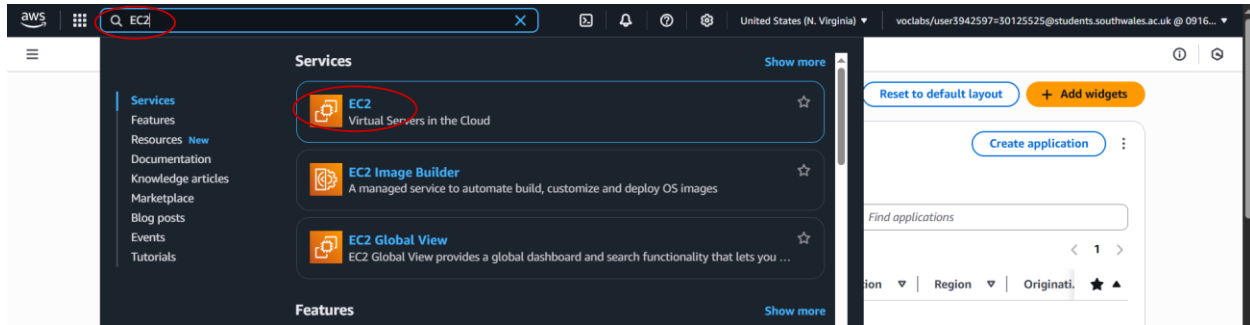


Figure 29. Navigating to EC2 Dashboard.

Step 21 – Launch a New Instance.

- Click **“Instances”** which is on the left panel and then click **“Launch Instances”** as depicted in figure 30.

This is to start the process of creating a new virtual machine.

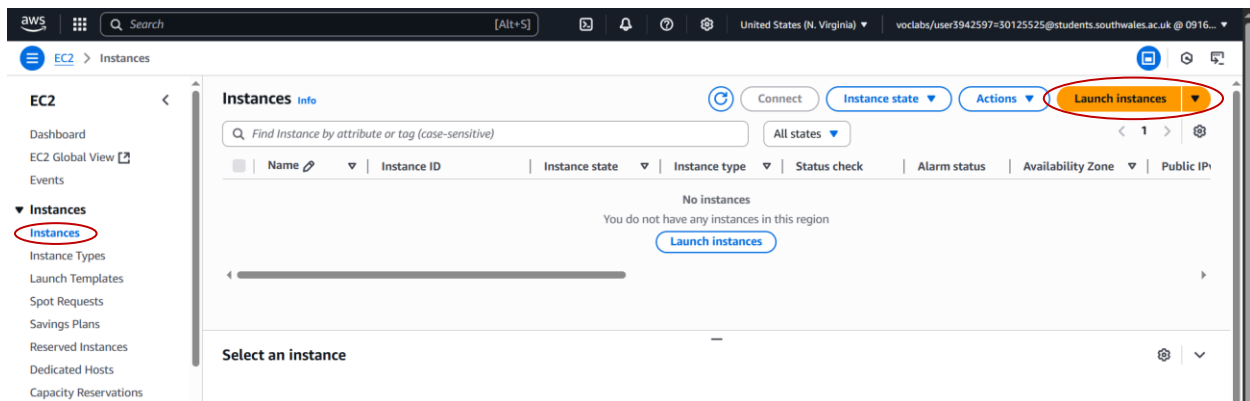


Figure 30. Launching an EC2 Instance.

Step 23 – Naming the instance.

- Under **“Name and tags”** give the desired name of your choice. Here, we will name our instance as **“30125525-datasafe-webserver”** as shown in figure 31.

The name should be unique for traceability.

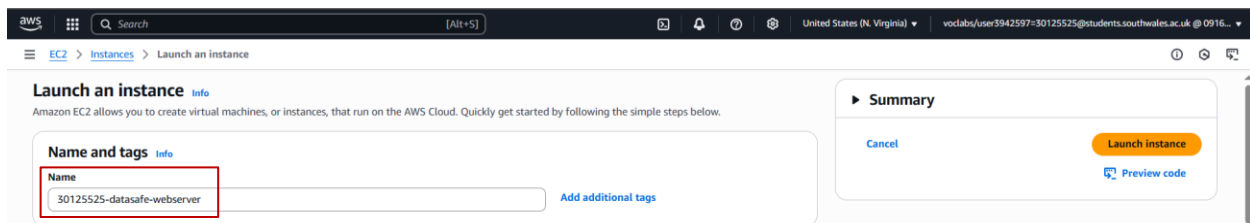


Figure 31. Naming our EC2 Instance.

Step 24 – Selecting AMI.

- Under “**Application and OS Images (Amazon Machine Image)**” select “**Amazon Linux 2023 AMI**” which is a free tier with a recent publish date as shown in figure 32.

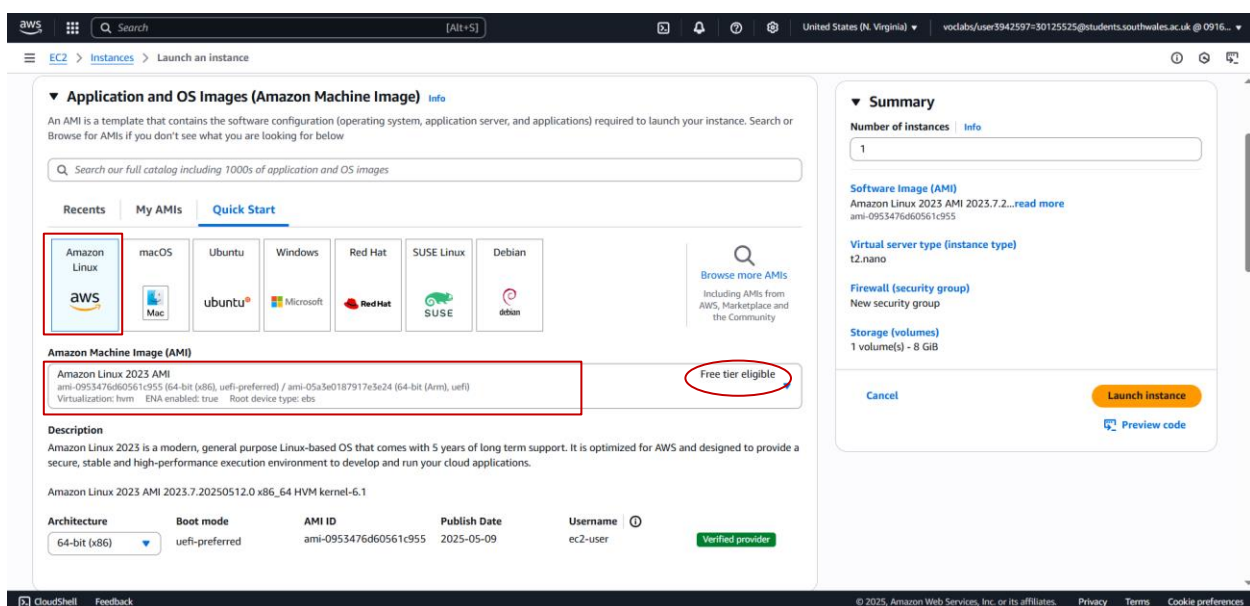


Figure 32. Selecting Ubuntu for AMI.

Step 25 – Choosing Instance Type.

- Under “**Instance Type**” select the best suited instance based on your budget and workload. Here, we will select “**t2.nano**” as shown below in figure 33.

We have opted for “t2.nano” since it is a low-cost, free tier eligible instance.

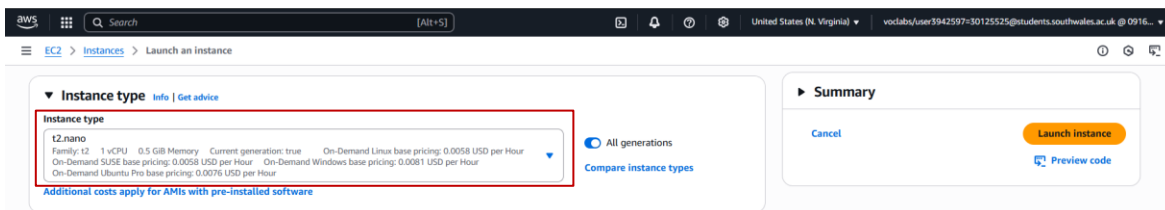


Figure 33. Choosing Instance Type as “t2.nano.”

Step 26 – Creating a Key Pair (Login)

- Under “**Key pair (login)**” click “**Create new key pair**” to create a key pair as shown in figure 34.

It is important to have a key pair so that we can use SSH access to the instance.

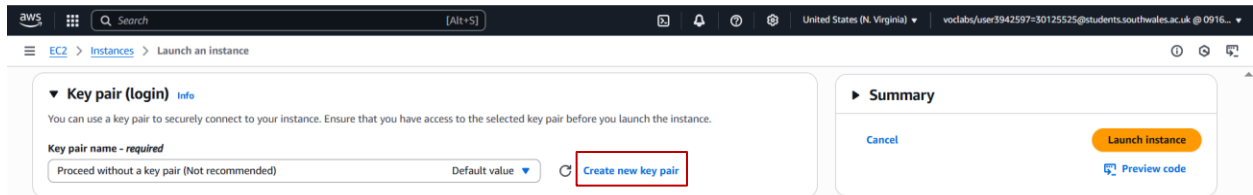


Figure 34. Creating a new key pair.

- Next, type in the desired key name. Here, we are naming our key pair as “**30125525-NewKey**”. Then, under “**Key pair type**” select “**RSA**.”
- For Private key file format select “**.pem**” and finally click “**Create key pair**” to create it as marked in figure 35.

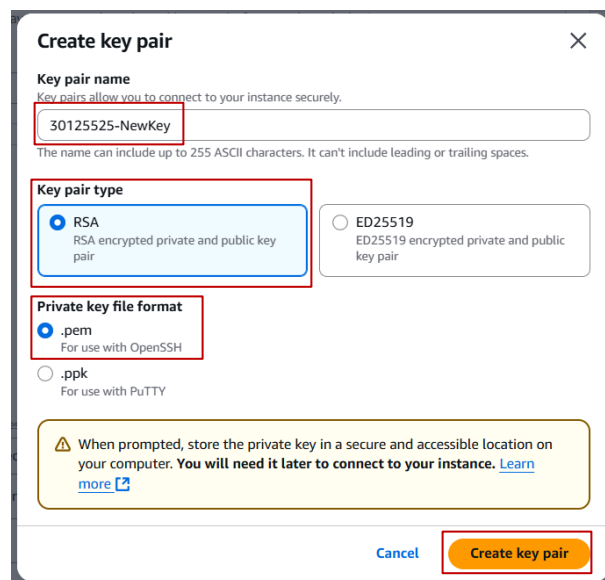


Figure 35. Finalizing key pair name and properties.

- After completing the above steps, a “**30125525-NewKey.pem**” file will be downloaded. Keep it securely as it will be needed later.

Step 27 – Configuring Network Setting.

- Click “**Edit**” and ensure that we have selected out VPC as shown in figure 36.
- Under “**Network Settings**” ensure that “**Auto-assign public IP**” is “**Enabled**” as shown in figure 36.

- Tick **“Allow SSH traffic from”** and IP as **“Anywhere”** for simplicity as shown in figure 36.
- Under **“Security group name”** put name as **“30125525-web-sg”**

Network settings Info

VPC - required Info
vpc-05a677218222a1482 (30125525-datasafe-vpc)

Subnet - Info
subnet-036920d1aaef88bc4 (30125525-public-subnet)
VPC: vpc-05a677218222a1482 Owner: 031661280974 Availability: Zone: us-east-1a
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.1.0/24

Auto-assign public IP Info
Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group Select existing security group

Security group name - required
30125525-web-sg
This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _./:/@.()+=&()!\$*

Description - required Info
launch-wizard-3 created 2025-05-27T20:44:05.893Z

Inbound Security Group Rules

Summary

Cancel Launch instance Preview code

Figure 36. Setting up Network Settings.

- Under **“Inbound Security Group Rules”** put **“Type”** as **“HTTP”** and **“Source Type”** to **“Anywhere”** as shown in figure 37.
- Add another security group rule. Under put **“Type”** as **“ssh”** and **“Source Type”** to **“Anywhere”** as shown in figure 37.

Inbound Security Group Rules

Security group rule 1 (TCP, 80, 0.0.0.0/0)

Type Info
HTTP

Source type Info
Anywhere

Protocol Info
TCP

Port range Info
80

Source Info
Add CIDR, prefix list or security group
0.0.0.0/0

Description - optional Info
e.g. SSH for admin desktop

Security group rule 2 (TCP, 22, 0.0.0.0/0)

Type Info
ssh

Source type Info
Anywhere

Protocol Info
TCP

Port range Info
22

Source Info
Add CIDR, prefix list or security group
0.0.0.0/0

Description - optional Info
e.g. SSH for admin desktop

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Add security group rule

Summary

Cancel Launch instance Preview code

Figure 37. Setting up Inbound Security Group Rules.

The HTTP rule allows public web traffic to access the application, while the SSH rule enables secure configuration.

Step 28 – Configuring Storage.

- Under “Configure storage” set the storage to “8 GiB” of “gp3” storage as shown in figure 38.

This is to enable general-purpose SSD volume type.

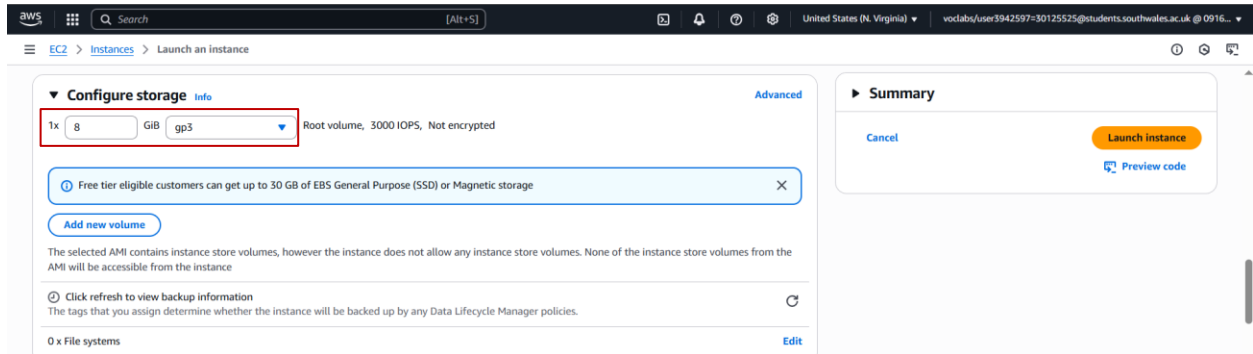


Figure 38. Setting up Storage.

Step 29. Enable Termination Protection.

- Under “Advanced details” enable “Termination Protection” as shown in below in figure 39.

This will be useful to prevent accidental termination of the instance.

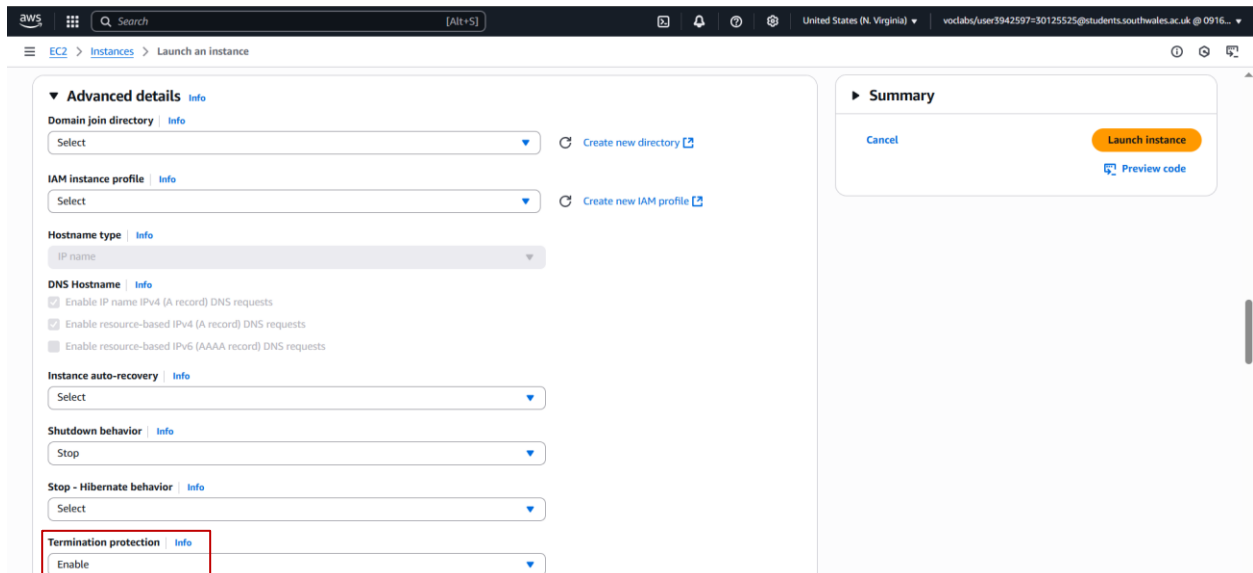
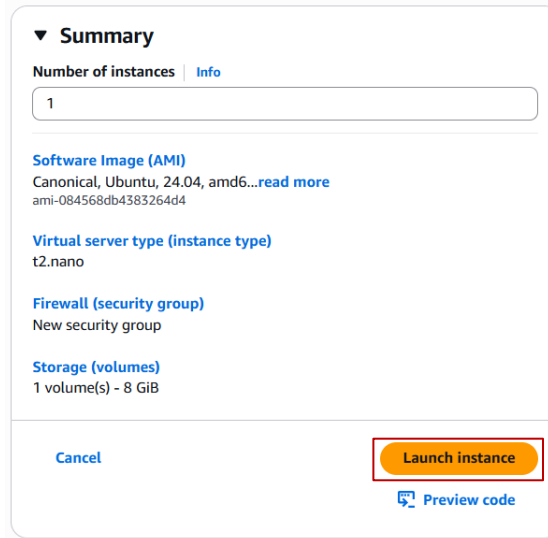


Figure 39. Enabling Termination Protection.

Step 30 – Launch Instance.

- Summarize and click “**Launch instance**” to create an EC2 instance as shown in figure 40.



The image shows the 'Summary' section of the AWS Management Console for launching an EC2 instance. It includes a 'Number of instances' input field set to 1, and sections for 'Software Image (AMI)', 'Virtual server type (instance type)', 'Firewall (security group)', and 'Storage (volumes)'. The 'Launch instance' button is highlighted with a red box.

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-084568db4383264d4

Virtual server type (instance type)
t2.nano

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Figure 40. Summarizing and launching the instance.

Step 31 – Connect to the EC2 Instance.

- Select the created instance and the click “**Connect**” as shown in the figure 41 below.

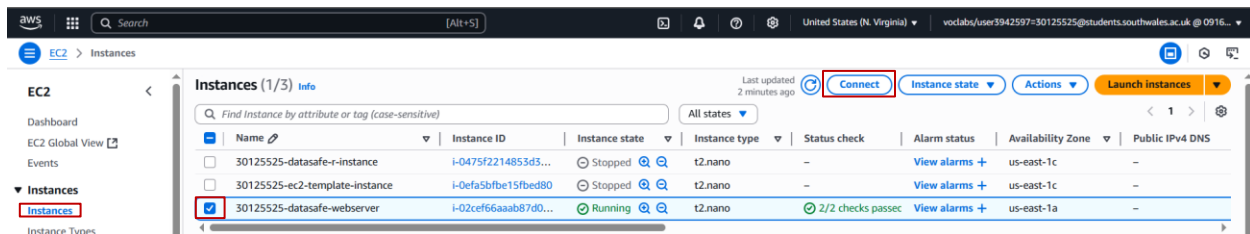


Figure 41. Connecting the EC2 Instance.

Step 32 – Launch and connecting to Terminal.

- Click “**Connect**” to launch the terminal shown in figure 42.

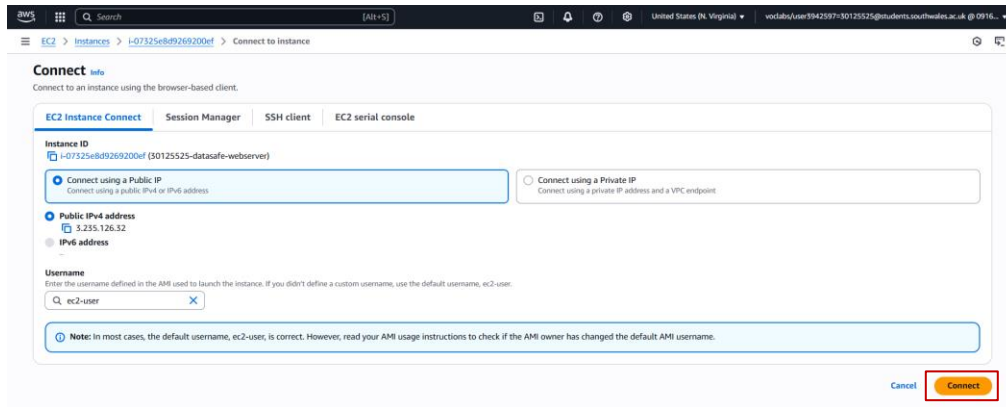


Figure 42. Connecting to the terminal.

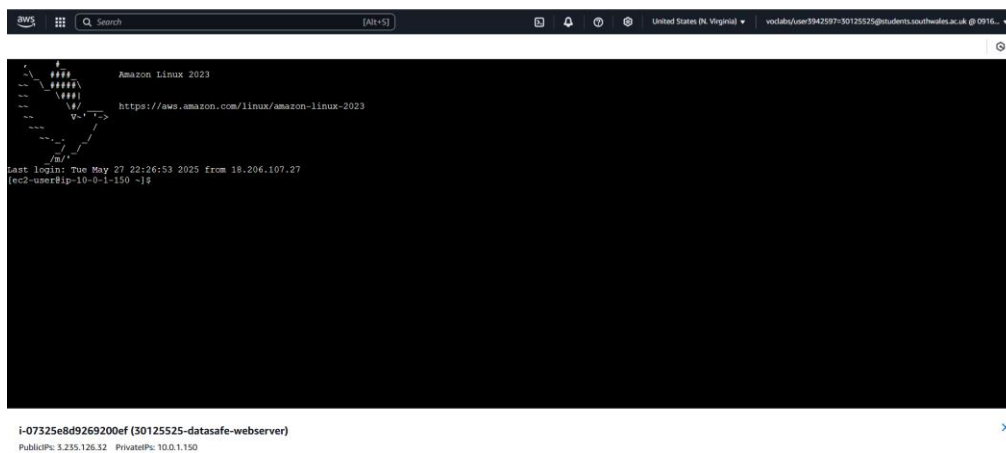


Figure 43. Connected to the terminal.

Step 33 – Gaining Root privileges.

- Run “**sudo su -**” as shown in figure 44.

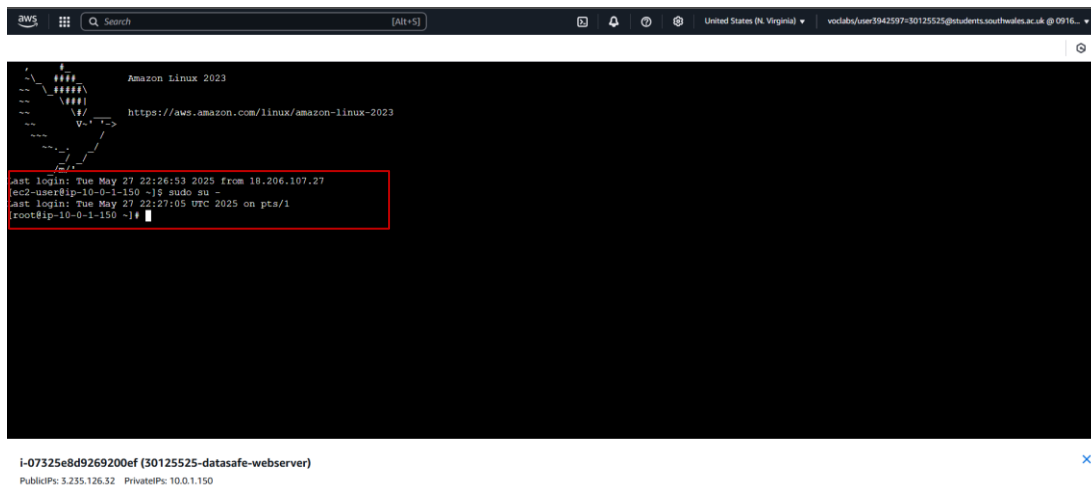
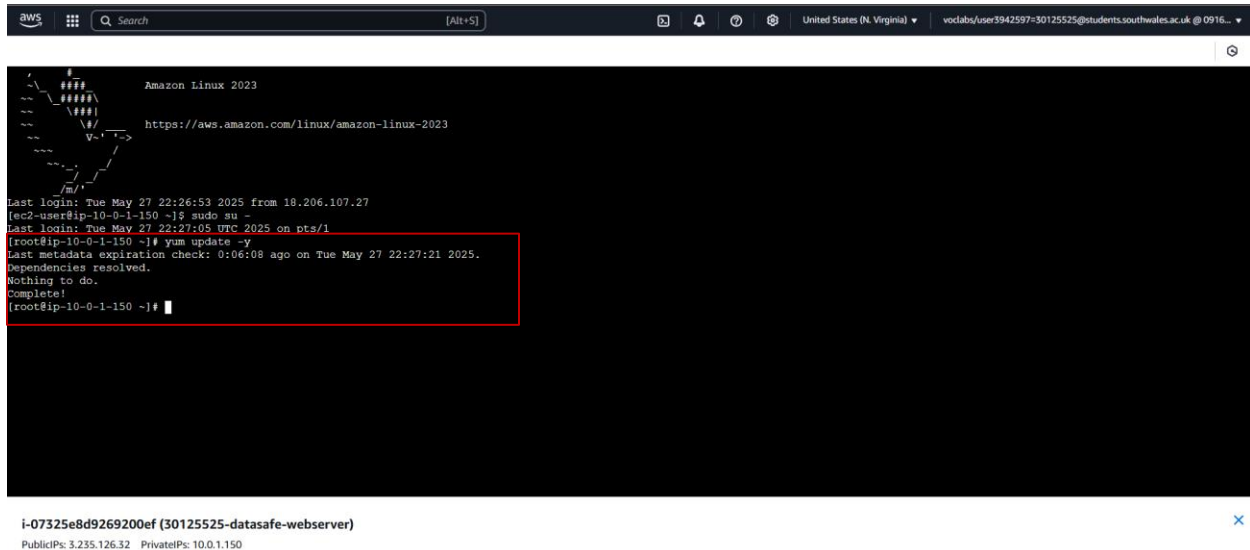


Figure 44. Running “sudo su -”

Step 34 – Updating the system.

- Running “yum update -y” as shown in figure 45.



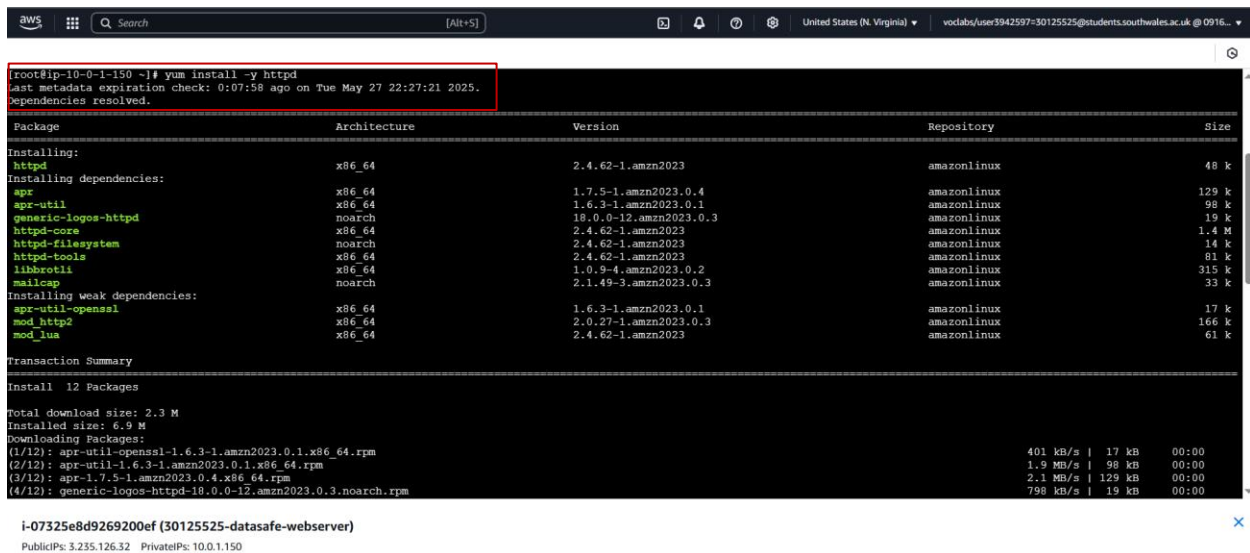
```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Tue May 27 22:26:53 2025 from 18.206.107.27
[ec2-user@ip-10-0-1-150 ~]$ sudo su -
Last login: Tue May 27 22:27:05 UTC 2025 on pts/1
[root@ip-10-0-1-150 ~]# yum update -y
Last metadata expiration check: 0:06:08 ago on Tue May 27 22:27:21 2025.
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-10-0-1-150 ~]#
```

Figure 45. Running Yum update command.

Step 35 – Installing Apache Web Server.

- Run “yum install -y httpd” as shown in the figure 46 below.



```
[root@ip-10-0-1-150 ~]# yum install -y httpd
Last metadata expiration check: 0:07:58 ago on Tue May 27 22:27:21 2025.
dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
httpd	x86_64	2.4.62-1.amzn2023	amazonlinux	48 k
Installing dependencies:				
apr	x86_64	1.7.5-1.amzn2023.0.4	amazonlinux	129 k
apr-util	x86_64	1.6.3-1.amzn2023.0.1	amazonlinux	98 k
generic-logos-httpd	noarch	18.0.0-12.amzn2023.0.3	amazonlinux	19 k
httpd-core	x86_64	2.4.62-1.amzn2023	amazonlinux	1.4 M
httpd-filesystem	noarch	2.4.62-1.amzn2023	amazonlinux	14 k
httpd-tools	x86_64	2.4.62-1.amzn2023	amazonlinux	81 k
libbrotli	x86_64	1.0.9-4.amzn2023.0.2	amazonlinux	315 k
mailcap	noarch	2.1.49-3.amzn2023.0.3	amazonlinux	33 k
Installing weak dependencies:				
apr-util-openssl	x86_64	1.6.3-1.amzn2023.0.1	amazonlinux	17 k
mod_http2	x86_64	2.0.27-1.amzn2023.0.3	amazonlinux	166 k
mod_lua	x86_64	2.4.62-1.amzn2023	amazonlinux	61 k

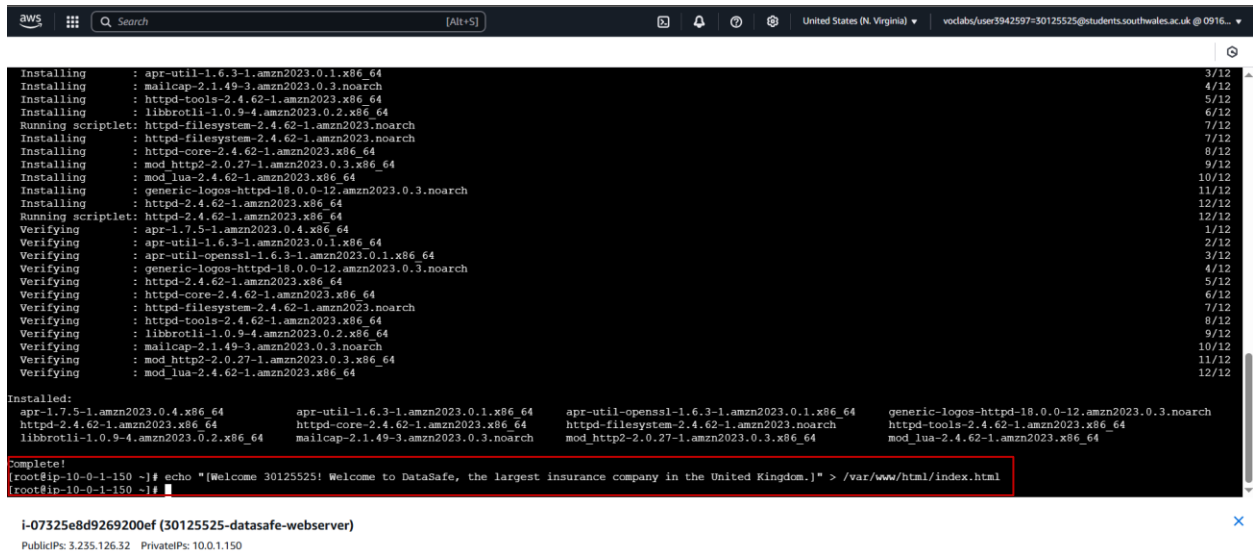
```
Transaction Summary
Install 12 Packages

Total download size: 2.3 M
Installed size: 6.9 M
Downloading Packages:
(1/12): apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64.rpm 401 kB/s | 17 kB 00:00
(2/12): apr-util-1.6.3-1.amzn2023.0.1.x86_64.rpm 1.9 MB/s | 98 kB 00:00
(3/12): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm 2.1 MB/s | 129 kB 00:00
(4/12): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm 798 kB/s | 19 kB 00:00
```

Figure 46. Installing Apache Web Server.

Step 36 – Creating the Web Page.

- Run “echo “[Welcome 30125525! Welcome to DataSafe, the largest insurance company in the United Kingdom.]” > /var/www/html/index.html”, shown in figure 47.



```
Installing      : apr-util-1.6.3-1.amzn2023.0.1.x86_64
Installing      : mailcap-2.1.49-3.amzn2023.0.3.noarch
Installing      : httpd-tools-2.4.62-1.amzn2023.x86_64
Installing      : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
Running scriptlet: httpd-filesystem-2.4.62-1.amzn2023.noarch
Installing      : httpd-filesystem-2.4.62-1.amzn2023.noarch
Installing      : httpd-core-2.4.62-1.amzn2023.x86_64
Installing      : mod_http2-2.0.27-1.amzn2023.0.3.x86_64
Installing      : mod_lua-2.4.62-1.amzn2023.x86_64
Installing      : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
Installing      : httpd-2.4.62-1.amzn2023.x86_64
Running scriptlet: httpd-2.4.62-1.amzn2023.x86_64
Verifying       : apr-1.7.5-1.amzn2023.0.4.x86_64
Verifying       : apr-util-1.6.3-1.amzn2023.0.1.x86_64
Verifying       : apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64
Verifying       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
Verifying       : httpd-2.4.62-1.amzn2023.x86_64
Verifying       : httpd-core-2.4.62-1.amzn2023.x86_64
Verifying       : httpd-filesystem-2.4.62-1.amzn2023.noarch
Verifying       : httpd-tools-2.4.62-1.amzn2023.x86_64
Verifying       : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
Verifying       : mailcap-2.1.49-3.amzn2023.0.3.noarch
Verifying       : mod_http2-2.0.27-1.amzn2023.0.3.x86_64
Verifying       : mod_lua-2.4.62-1.amzn2023.x86_64

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64      apr-util-1.6.3-1.amzn2023.0.1.x86_64      apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64      generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
httpd-2.4.62-1.amzn2023.x86_64      httpd-core-2.4.62-1.amzn2023.x86_64      httpd-filesystem-2.4.62-1.amzn2023.noarch      httpd-tools-2.4.62-1.amzn2023.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64  mailcap-2.1.49-3.amzn2023.0.3.noarch      mod_http2-2.0.27-1.amzn2023.0.3.x86_64      mod_lua-2.4.62-1.amzn2023.x86_64

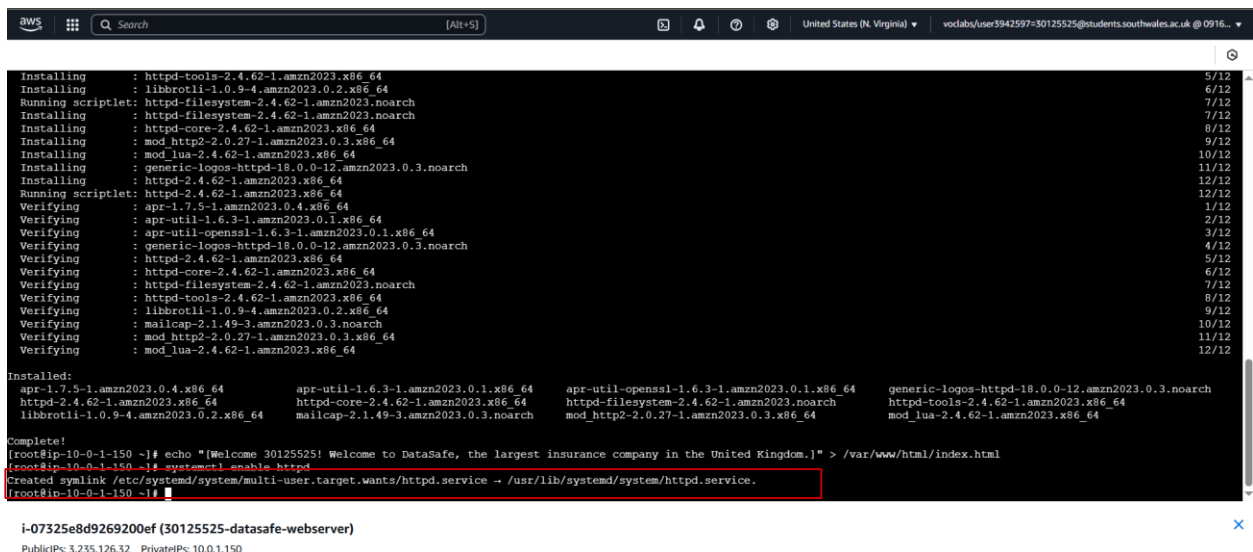
Complete!
[root@ip-10-0-1-150 ~]# echo "[Welcome 30125525! Welcome to DataSafe, the largest insurance company in the United Kingdom.]" > /var/www/html/index.html
[root@ip-10-0-1-150 ~]#
```

i-07325e8d9269200ef (30125525-datasafe-webserver)
PublicIPs: 3.235.126.32 PrivateIPs: 10.0.1.150

Figure 47. Creating an HTML file with some content.

Step 37 – Enabling Apache.

- Run “systemctl enable httpd” as shown in figure 48 below.



```
Installing      : httpd-tools-2.4.62-1.amzn2023.x86_64
Installing      : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
Running scriptlet: httpd-filesystem-2.4.62-1.amzn2023.noarch
Installing      : httpd-filesystem-2.4.62-1.amzn2023.noarch
Installing      : httpd-core-2.4.62-1.amzn2023.x86_64
Installing      : mod_http2-2.0.27-1.amzn2023.0.3.x86_64
Installing      : mod_lua-2.4.62-1.amzn2023.x86_64
Installing      : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
Installing      : httpd-2.4.62-1.amzn2023.x86_64
Running scriptlet: httpd-2.4.62-1.amzn2023.x86_64
Verifying       : apr-1.7.5-1.amzn2023.0.4.x86_64
Verifying       : apr-util-1.6.3-1.amzn2023.0.1.x86_64
Verifying       : apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64
Verifying       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
Verifying       : httpd-2.4.62-1.amzn2023.x86_64
Verifying       : httpd-core-2.4.62-1.amzn2023.x86_64
Verifying       : httpd-filesystem-2.4.62-1.amzn2023.noarch
Verifying       : httpd-tools-2.4.62-1.amzn2023.x86_64
Verifying       : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
Verifying       : mailcap-2.1.49-3.amzn2023.0.3.noarch
Verifying       : mod_http2-2.0.27-1.amzn2023.0.3.x86_64
Verifying       : mod_lua-2.4.62-1.amzn2023.x86_64

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64      apr-util-1.6.3-1.amzn2023.0.1.x86_64      apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64      generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
httpd-2.4.62-1.amzn2023.x86_64      httpd-core-2.4.62-1.amzn2023.x86_64      httpd-filesystem-2.4.62-1.amzn2023.noarch      httpd-tools-2.4.62-1.amzn2023.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64  mailcap-2.1.49-3.amzn2023.0.3.noarch      mod_http2-2.0.27-1.amzn2023.0.3.x86_64      mod_lua-2.4.62-1.amzn2023.x86_64

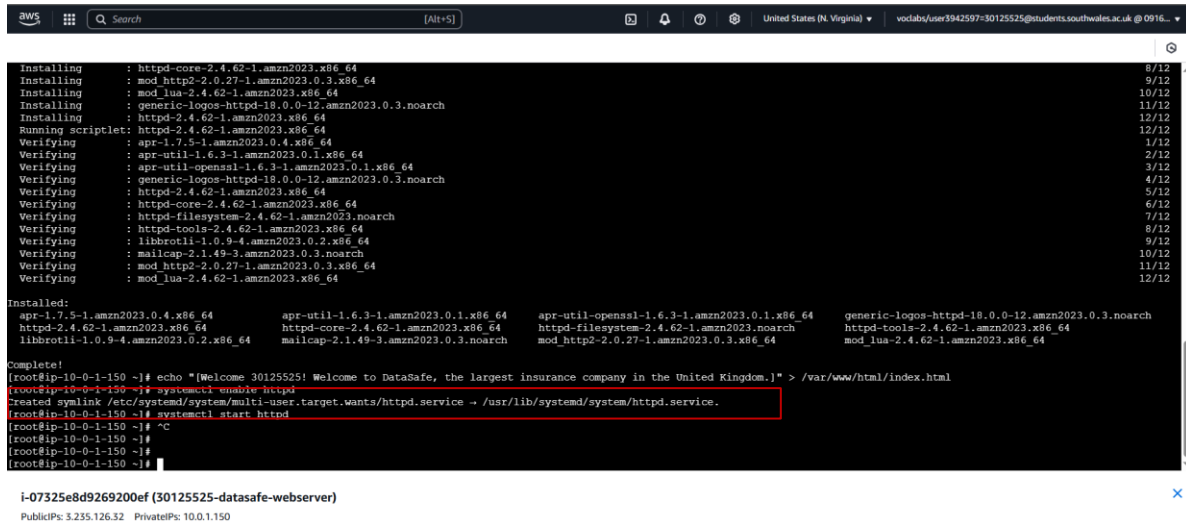
Complete!
[root@ip-10-0-1-150 ~]# echo "[Welcome 30125525! Welcome to DataSafe, the largest insurance company in the United Kingdom.]" > /var/www/html/index.html
[root@ip-10-0-1-150 ~]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-10-0-1-150 ~]#
```

i-07325e8d9269200ef (30125525-datasafe-webserver)
PublicIPs: 3.235.126.32 PrivateIPs: 10.0.1.150

Figure 48. Enabling Apache.

Step 38 – Start Apache.

- Run “systemctl start httpd” as shown in figure 49.



```
Installing : httpd-core-2.4.62-1.amzn2023.x86_64
Installing : mod_http2-2.0.27-1.amzn2023.0.3.x86_64
Installing : mod_lua-2.4.62-1.amzn2023.x86_64
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
Installing : httpd-2.4.62-1.amzn2023.x86_64
Running scriptlet: httpd-2.4.62-1.amzn2023.x86_64
Verifying : apr-1.7.5-1.amzn2023.0.4.x86_64
Verifying : apr-util-1.6.3-1.amzn2023.0.1.x86_64
Verifying : apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
Verifying : httpd-2.4.62-1.amzn2023.x86_64
Verifying : httpd-core-2.4.62-1.amzn2023.x86_64
Verifying : httpd-filesystem-2.4.62-1.amzn2023.noarch
Verifying : httpd-tools-2.4.62-1.amzn2023.x86_64
Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch
Verifying : mod_http2-2.0.27-1.amzn2023.0.3.x86_64
Verifying : mod_lua-2.4.62-1.amzn2023.x86_64

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64      apr-util-1.6.3-1.amzn2023.0.1.x86_64  apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
httpd-2.4.62-1.amzn2023.x86_64      httpd-core-2.4.62-1.amzn2023.x86_64  httpd-filesystem-2.4.62-1.amzn2023.noarch      httpd-tools-2.4.62-1.amzn2023.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64  mailcap-2.1.49-3.amzn2023.0.3.noarch  mod_http2-2.0.27-1.amzn2023.0.3.x86_64      mod_lua-2.4.62-1.amzn2023.x86_64

Complete!
[root@ip-10-0-1-150 ~]# echo "[Welcome 30125525! Welcome to DataSafe, the largest insurance company in the United Kingdom.]" > /var/www/html/index.html
[root@ip-10-0-1-150 ~]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-10-0-1-150 ~]# systemctl start httpd
[root@ip-10-0-1-150 ~]# ^C
[root@ip-10-0-1-150 ~]#
[root@ip-10-0-1-150 ~]#
```

Figure 49. Running our Web Page.

Step 39 – Validating our Web Page.

- Copy the instance’s “Public IPv4 address” as shown in figure 50.
- Open a new tab and type `http://<your copied IP>`. Here, my link is “`http://3.235.126.32`.”

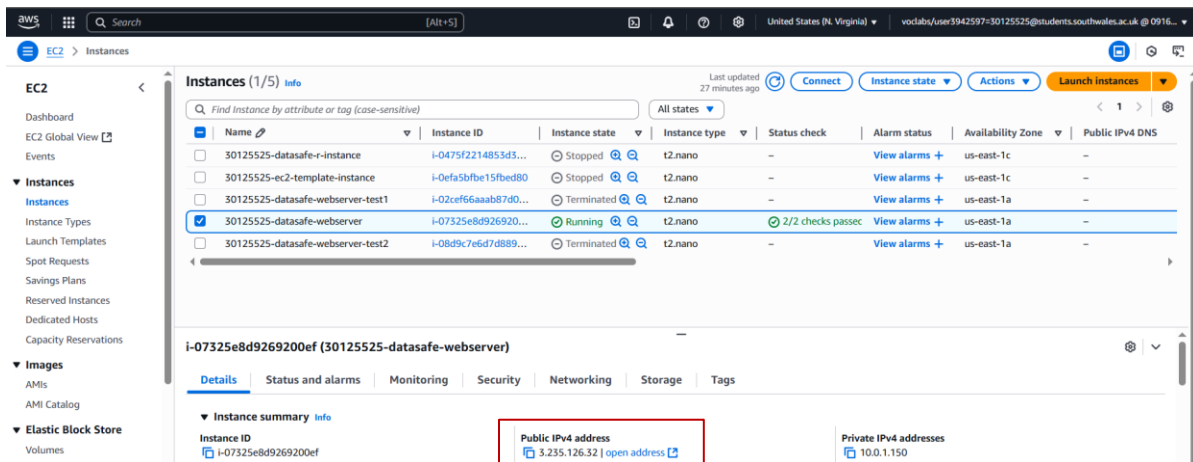


Figure 50. Copying the instance's IPv4 address.

- In figure 51, we can see that the message is printed successfully and is publicly accessible.



Figure 51. Web Page with the desired content.

Step 40 – Stop EC2 Instance.

- Ensure to stop **unrequired** running instances after completing the work. Doing this will save us a lot of money.
- Select the instance you want to stop and click “**Instance State**” and then click “**Stop Instance**” as shown in figure 52.

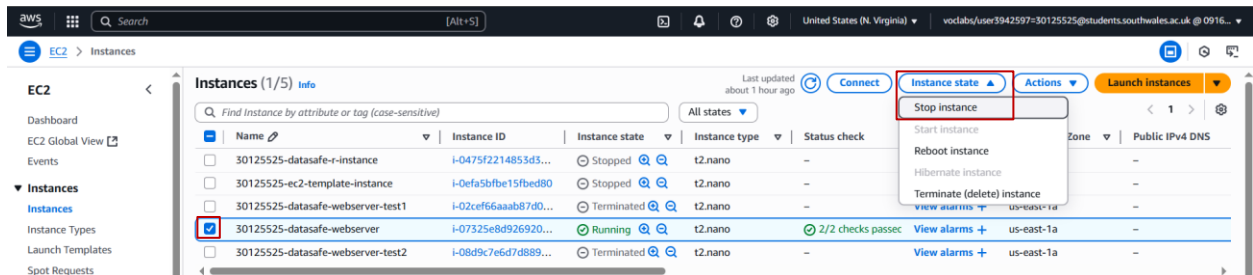


Figure 52. Stopping unrequired instances.

References:

[1] Amazon Web Services, "Amazon Virtual Private Cloud (VPC) User Guide," Amazon Web Services, 2025. [Online]. Available:

<https://docs.aws.amazon.com/vpc/latest/userguide/what-is-amazon-vpc.html>.

[Accessed: 26 May 2025].

[2] Amazon Web Services, "Amazon EC2 User Guide for Linux Instances," Amazon Web Services, 2025. [Online]. Available:

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/concepts.html>. [Accessed: 27

May 2025].

[3]: Amazon Web Services, "AWS Security Groups for Your VPC," Amazon Web Services, 2025. [Online]. Available:

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_SecurityGroups.html.

[Accessed: 27 May 2025].